

TESTs - Lesson 06 - Building Unit Tests

When initializing Hardhat, which project type does the instructor recommend for this lesson?



- A Create a JavaScript project
- B Create a TypeScript project (with Viem)**
- C Create an empty hardhat.config.js

What does "HRE" stand for in the context of Hardhat?



- A Hardhat Remote Environment
- B Hardhat Runtime Evolution
- C Hardhat Runtime Environment**
- D Holy Roman Empire

What is a better way of understanding the behavior of a complex smart contract like 'Lock.sol' more easily than reading its Solidity code?



- A By directly deploying it to a testnet and observing its live interactions.
- B By consulting external documentation and whitepapers.
- C By analyzing the human-readable descriptions generated by its unit tests.**
- D By refactoring the contract to simplify its structure.

What is Hardhat's role described as, akin to a "sandbox"?



- A A platform for secure cryptocurrency transactions.
- B An environment for testing smart contracts freely.**
- C A tool for compiling Solidity code into bytecode.

Which of the following is **NOT** a component or tool introduced or extensively discussed in this lesson for smart contract testing?



A Hardhat Runtime Environment (HRE)

B Viem library

C Ganache

D Mocha and Chai

What is the primary benefit of using a testing methodology like TDD for smart contract development?



A It allows for rapid iteration and quick deployment of contracts.

B It ensures that contracts are fully functional before any cost-incurring deployments due to the irreversibility of blockchain.

C It simplifies the process of writing complex Solidity code.

Why should you use parallel testing environments using tools like Hardhat and Mocha?



A To speed up test execution by running multiple tests simultaneously.

B To ensure that each test runs in isolation, preventing one test from affecting the state or outcome of another.

C To allow for easier integration with front-end applications.

What is the key advantage of using a debugger (like the one in VS Code with breakpoints) during smart contract testing?



A It automates the entire testing process without manual intervention.

B It allows developers to pause execution, inspect variable values, and control flow to understand contract behavior.

C It guarantees the contract is free of security vulnerabilities.

In today's lesson a change in the `HelloWorld.sol` contract's constructor text immediately caused a test to fail. What core principle of robust software development does this demonstrate?



- A** The importance of code refactoring.
- B** The necessity of continuous integration.
- C** The value of automated testing for validating expected behavior.